

Program : Diploma in Instrumentation Engineering	
Course Code : 3087	Course Title: Analog Circuits Lab for Instrumentation
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide experience on constructing CE and CC amplifiers.
- To provide experience on building various oscillator circuits.
- To design various application circuits of Operational Amplifier.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic of Electronics and Electrical		Electronic Instrumentation	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Construct amplifier in CE and CC configuration.	6	Applying
CO2	Build various oscillator circuits	9	Applying
CO3	Design and implement linear circuits using operational amplifier	18	Applying
CO4	Design and implement non-linear circuits using operational amplifier	6	Applying
	Lab Test	6	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3					2	
CO2	3					2	
CO3	3					2	2
CO4	3					2	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Construct amplifier in CE and CC configuration		
M1.01	Construct a single stage RC Coupled CE amplifier, observe its phase difference between input and output waveforms, and plot its frequency response.	3	Applying
M1.02	Construct an emitter follower circuit and measure its gain.	3	Applying
CO2	Build various oscillator circuits		
M2.01	Build an RC phase shift oscillator	3	Applying
M2.02	Build a Colpitt's oscillator	3	Applying
M2.03	Construct a BJT astable multivibrator and plot the waveforms at base and collector of transistor.	3	Applying
	Lab Test – I	3	
CO3	Design and implement linear circuits using operational amplifier		
M3.01	Determine Operational Amplifier parameters: input bias current, input offset voltage, output offset voltage, CMRR and slew rate.	3	Applying
M3.02	Design and set up an inverting and non-inverting amplifier using Op- amp	3	Applying
M3.03	Design and set up integrator and differentiator circuit using Op amp	3	Applying

M3.04	Design and set up current to voltage and voltage to current converter using Op amp	3	Applying
M3.05	Design an adder and subtractor using Op amp	3	Applying
M3.06	Design an instrumentation amplifier and find the CMRR.	3	Applying
CO4	Design and implement non-linear circuits using operational amplifier		
M4.01	Design and set up Schmitt trigger circuit using Op amp.	1.5	Applying
M4.02	Set up a Precision rectifier using Op amp.	1.5	Applying
M4.03	Design and set up various clipping circuits using Op amp.	1.5	Applying
M4.04	Design and set up various clamping circuits using Op amp	1.5	Applying
	Lab Test – II	3	

Text / Reference:

T/R	Book Title/Author
R1	David Buchla, Lab Manual for Electronics Fundamentals and Electronic Circuits Fundamentals, Pearson College Publisher.
R2	James M Fiore, Operational Amplifiers & Linear Integrated Circuits: Theory and Application Laboratory Manual/3E, Delmar, Thomson Learning Publisher.
R3	Lab Manual Vol. I – K. Navas, PHI Publisher.
R4	Lab Manual Vol. II- K Navas, PHI Publisher.

Online Resources:

Sl.No	Website Link
1	https://www.scribd.com/document/91915901/ECE-Lab-Manual
2	scitechpublications.com
3	edutalks.org
4	https://www.chegg.com
5	https://www2.mvcc.edu/users/faculty/jfiore/OpAmps/labs/Laboratory